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DECIMALS

1. Decimal Fractions

Decimal fractions are fractions where the denominator is a power of 10, like 10, 100, or 1000. Instead of writing these fractions as numbers over another number, we can use decimals.

Examples:

- 1/10 can be written as 0.1 (zero point one).
- 3/100 can be written as 0.03 (zero point zero three).
- 5/1000 can be written as 0.005 (zero point zero zero five).

- **Write down the decimal form of the following fractions.**

1) $\frac{7}{10}$ = ______	2) $\frac{25}{100}$ = ______	3) $\frac{40}{1000}$ = ______

- **Convert these fractions into decimals.**

1) $\frac{2}{10}$ = ______	2) $\frac{9}{100}$ = ______	3) $\frac{35}{1000}$ = ______

2. Tenths, Hundredths, Thousandths

Decimals have place values, just like whole numbers. Each place after the decimal point represents a fraction of a whole.

- **Tenths (0.1):** The first digit to the right of the decimal point.
- **Hundredths (0.01):** The second digit to the right of the decimal point.
- **Thousandths (0.001):** The third digit to the right of the decimal point.

- **Write down the decimal form of the following fractions.**

1) $\frac{6}{10}$ = ______

2) $\frac{4}{100}$ = ______

3) $\frac{789}{1000}$ = ______

3. Place Value Chart for Decimals

Place value helps us understand the value of each digit in a decimal number.

Hundreds	Tens	Units	.	Tenths	Hundredths	Thousandths
1	2	3	.	4	5	6

- **Use a place value chart to write these numbers:**

1) 3.678

Hundreds	Tens	Units	.	Tenths	Hundredths	Thousandths

2) 45.09

Hundreds	Tens	Units	.	Tenths	Hundredths	Thousandths

3) 0.304

Hundreds	Tens	Units	.	Tenths	Hundredths	Thousandths

4. Adding Decimals

Decimals can be added just like whole numbers. Align the decimal points vertically before adding.

Example: $4.9 + 0.4 = 5.3$

- **Exercise:**

1) $0.6 = \frac{6}{10}$

2) $4.9 =$ ____

- **Write as decimals and fractions.**

1) Three-tenths	

2) Seven-tenths	

3) Five and two-tenths	

4) Two and three-tenths	

5) Eight-tenths	

6) Seven and six-tenths	

7) Nine and five-tenths	

8) Seven and one-tenth	

- **Write these numbers in words.**

1) 25.9 =

2) 314.4 =

3) 42.3 =

4) 89.7 =

5) 150.5 =

- **Convert the following mixed fractions to decimal form.**

1) $5\frac{4}{10}$ = ____________

2) $2\frac{7}{10}$ = ____________

3) $8\frac{1}{5}$ = ____________

4) $6\frac{3}{4}$ = ____________

5) $4\frac{9}{10}$ = ____________

- **Mixed Fractions and Decimals**

A mixed fraction combines a whole number and a fraction. We can convert it to a decimal by writing the fraction as a decimal.

Example: $2\frac{3}{10} = 2.3$

- **Convert these mixed fractions to decimals**

1) $5\frac{6}{10}$ = ____________

2) $3\frac{25}{100}$ = ____________

- **Write these decimals as mixed fractions**

1) 3.25 = ___

2) 9.6 = ___

3) 15.75 = ___

4) 5.4 = ___

5) 11.08 = ___

- **Addition and Subtraction of Decimals (Up to 2 and 3 Decimal Places)**

When adding or subtracting decimals, make sure to align the decimal points vertically before performing the calculations. This ensures each place value is correctly aligned (tenths, hundredths, thousandths).

Examples:

1. Addition Example:

- 5.25 + 3.75
- Align the decimals:

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  5.25
+ 3.75
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  9.00
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- **Answer:** 9.00

2. Subtraction Example:

- 10.6 - 2.34
- Align the decimals:

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$$\begin{array}{r} 10.60 \\ -\ 2.34 \\ \hline 8.26 \\ \hline \end{array}$$

- **Answer:** 8.26

- **Add the following.**

$$\begin{array}{r} 12.30 \\ +\ 7.50 \\ \hline \\ \hline \end{array} \qquad \begin{array}{r} 5.80 \\ +\ 2.40 \\ \hline \\ \hline \end{array} \qquad \begin{array}{r} 6.70 \\ +\ 3.20 \\ \hline \\ \hline \end{array} \qquad \begin{array}{r} 3.23 \\ +\ 0.93 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 1.37 \\ +\ 0.65 \\ \hline \\ \hline \end{array} \qquad \begin{array}{r} 9.99 \\ +\ 0.33 \\ \hline \\ \hline \end{array}$$

- **Subtract the following.**

$$\begin{array}{r} 7.65 \\ -\ 2.34 \\ \hline \\ \hline \end{array} \qquad \begin{array}{r} 5.25 \\ -\ 0.11 \\ \hline \\ \hline \end{array} \qquad \begin{array}{r} 3.02 \\ -\ 2.01 \\ \hline \\ \hline \end{array} \qquad \begin{array}{r} 8.64 \\ -\ 3.34 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 6.65 \\ -\ 4.04 \\ \hline \\ \hline \end{array} \qquad \begin{array}{r} 9.65 \\ -\ 5.22 \\ \hline \\ \hline \end{array}$$